THE FUNCTOR OF POINTS WHERE A MODULE IS FREE AT THE GENERIC POINTS OF THE FIBRES IS AN FPQC SHEAF AND LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source line 6267, 'Some details omitted.', is retained verbatim at the end of the counted proof, as is the author's own footnote attached to the fpqc covering in the first line of the proof. The sheaf argument and the limit preserving reduction are both carried out in the span; only the repeated verification of the same reduction is left to the reader.

Allowed standard prerequisites (editorial).

- The situation defining the functor of points where the module is free of a fixed rank at the generic point of the fibre. Same chapter flat.tex, situation and equation labelled situation-free-at-generic-points and equation-free-at-generic-point-fibre (statement, proof line 6180; used at proof lines 6197, 6241 and 6266). Fixes what H_p is and what condition on the pullback of F the proof has to establish; the two parts of the proof are then organised around that condition.
- Descent of finite projective modules along a flat local homomorphism, and generation of a finite type module on a neighbourhood. Algebra, label algebra-lemma-finite-projective-descends (proof line 6209); Modules, label modules-lemma-finite-type-surjective-on-stalk (proof line 6213). The first propagates the local freeness of the pullback to the limit point through the flat local map of stalks; the second produces an affine neighbourhood of the generic point and a surjection from a free module onto the restriction of the module.
- The induction step lemma and the openness of flatness, with the limit descent lemmas. Same chapter flat.tex, label lemma-induction-step (proof

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line 6225); More on Morphisms, label more-morphisms-theorem-openness-flatness (proof line 6246); Limits, labels limits-lemma-descend-opens (proof line 6251) and limits-lemma-descend-module-flat-finite-presentation (proof line 6255). The induction step converts the isomorphism on every fibre into injectivity after tensoring the global sections with the local rings, which upgrades the displayed surjection to an isomorphism; openness of flatness and the two descent lemmas reduce the limit preserving question to a stage of the inverse system.

Source reference. Original lemma 00O1: sources/stacks-chow/tags.txt, source lines 879–879. Navigation pointer for \ref{algebra-lemma-finite-projective-descends} used inside the counted span (line 6209); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MQ: sources/stacks-flat/pinned-original-source.tex, source lines 6160–6160. Navigation pointer for \ref{equation-free-at-generic-point-fibre} used inside the counted span (line 6197); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I3: sources/stacks-flat/pinned-original-source.tex, source lines 2356–2356. Navigation pointer for \ref{lemma-induction-step} used inside the counted span (line 6225); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LY: sources/stacks-chow/tags.txt, source lines 7252–7252. Navigation pointer for \ref{limits-lemma-descend-module-flat-finite-presentation} used inside the counted span (line 6255); the argument that consumes it is retained in the span.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Navigation pointer for \ref{limits-lemma-descend-opens} used inside the counted span (line 6251); the argument that consumes it is retained in the span.

Source reference. Original lemma 01B8: sources/stacks-chow/tags.txt, source lines 1717–1717. Navigation pointer for \ref{modules-lemma-finite-type-surjective-on-stalk} used inside the counted span (line 6213); the argument that consumes it is retained in the span.

Source reference. Original lemma 0399: sources/stacks-chow/tags.txt, source lines 4264–4264. Navigation pointer for \ref{more-morphisms-theorem-openness-flatness} used inside the counted span (line 6246); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MP: sources/stacks-flat/pinned-original-source.tex, source lines 6146–6146. Navigation pointer for \ref{situation-free-at-generic-points} used inside the counted span (line 6180); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05MP.*

(1) *The functor H_p satisfies the sheaf property for the fpqc topology.*
(2) *If $\mathcal{F}$ is of finite presentation, then functor H_p is limit preserving.*

Proof. Let $\{T_i \to T\}_{i \in I}$ be an fpqc[1] covering of schemes over S. Set $X_i = X_{T_i} = X \times_S T_i$ and denote $\mathcal{F}_i$ the pullback of $\mathcal{F}$ to X_i. Assume that $\mathcal{F}_i$ satisfies (05MQ)

[1]It is quite easy to show that H_p is a sheaf for the fppf topology using that flat morphisms of finite presentation are open. This is all we really need later on. But it is kind of fun to prove directly that it also satisfies the sheaf condition for the fpqc topology.

for all i. Pick $t \in T$ and let $\xi_t \in X_T$ denote the generic point of X_t. We have to show that $\mathcal{F}$ is free in a neighbourhood of ξ_t. For some $i \in I$ we can find a $t_i \in T_i$ mapping to t. Let $\xi_i \in X_i$ denote the generic point of X_{t_i}, so that ξ_i maps to ξ_t. The fact that $\mathcal{F}_i$ is free of rank p in a neighbourhood of ξ_i implies that $(\mathcal{F}_i)_{x_i} \cong \mathcal{O}_{X_i,x_i}^{\oplus p}$ which implies that $\mathcal{F}_{T,\xi_t} \cong \mathcal{O}_{X_T,\xi_t}^{\oplus p}$ as $\mathcal{O}_{X_T,\xi_t} \to \mathcal{O}_{X_i,x_i}$ is flat, see for example Algebra, Lemma 00O1. Thus there exists an affine neighbourhood U of ξ_t in X_T and a surjection $\mathcal{O}_U^{\oplus p} \to \mathcal{F}_U = \mathcal{F}_T|_U$, see Modules, Lemma 01B8. After shrinking T we may assume that $U \to T$ is surjective. Hence $U \to T$ is a smooth morphism of affines with geometrically irreducible fibres. Moreover, for every $t' \in T$ we see that the induced map

$$\alpha : \mathcal{O}_{U,\xi_{t'}}^{\oplus p} \longrightarrow \mathcal{F}_{U,\xi_{t'}}$$

is an isomorphism (since by the same argument as before the module on the right is free of rank p). It follows from Lemma 05I3 that

$$\Gamma(U, \mathcal{O}_U^{\oplus p}) \otimes_{\Gamma(T,\mathcal{O}_T)} \mathcal{O}_{T,t'} \longrightarrow \Gamma(U, \mathcal{F}_U) \otimes_{\Gamma(T,\mathcal{O}_T)} \mathcal{O}_{T,t'}$$

is injective for every $t' \in T$. Hence we see the surjection α is an isomorphism. This finishes the proof of (1).

Assume that $\mathcal{F}$ is of finite presentation. Let $T = \lim_{i \in I} T_i$ be a directed limit of affine S-schemes and assume that $\mathcal{F}_T$ satisfies (05MQ). Set $X_i = X_{T_i} = X \times_S T_i$ and denote $\mathcal{F}_i$ the pullback of $\mathcal{F}$ to X_i. Let $U \subset X_T$ denote the open subscheme of points where $\mathcal{F}_T$ is flat over T, see More on Morphisms, Theorem 0399. By assumption every generic point of every fibre is a point of U, i.e., $U \to T$ is a smooth surjective morphism with geometrically irreducible fibres. We may shrink U a bit and assume that U is quasi-compact. Using Limits, Lemma 01Z4 we can find an $i \in I$ and a quasi-compact open $U_i \subset X_i$ whose inverse image in X_T is U. After increasing i we may assume that $\mathcal{F}_i|_{U_i}$ is flat over T_i, see Limits, Lemma 05LY. In particular, $\mathcal{F}_i|_{U_i}$ is finite locally free hence defines a locally constant rank function $\rho : U_i \to \{0, 1, 2, \ldots\}$. Let $(U_i)_p \subset U_i$ denote the open and closed subset where ρ has value p. Let $V_i \subset T_i$ be the image of $(U_i)_p$; note that V_i is open and quasi-compact. By assumption the image of $T \to T_i$ is contained in V_i. Hence there exists an $i' \geq i$ such that $T_{i'} \to T_i$ factors through V_i by Limits, Lemma 01Z4. Then $\mathcal{F}_{i'}$ satisfies (05MQ) as desired. Some details omitted. $\square$